



PT SAPTAINDRA SEJATI

FINAL DRIVE RH
RECEIVING INSPECTION
CHECKSHEET

MACHINE MODEL

HD785-7

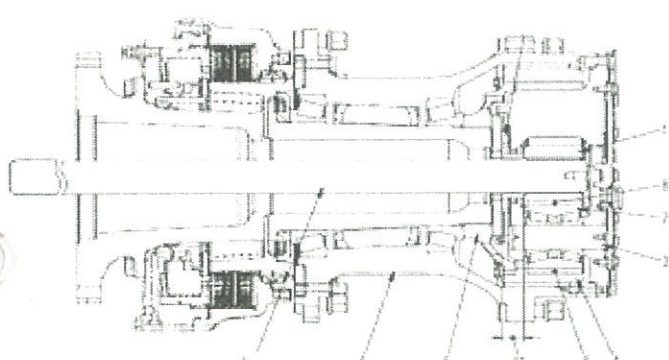
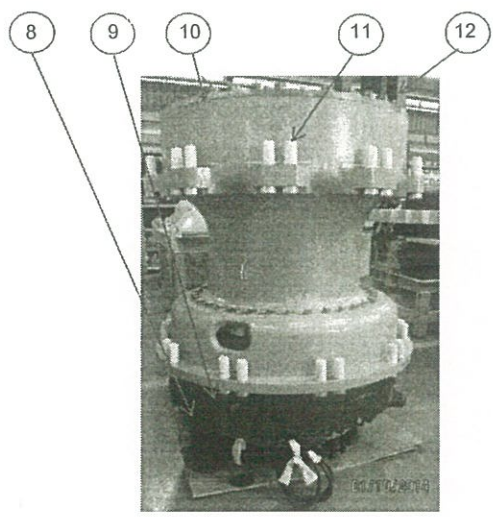
RO NO :

POWERTRAIN SECTION

RECEIVING INSPECTION SHEET FINAL DRIVE	MACHINE MODEL	REPORT NO.	DOCUMENT NO.	QR01-PT-027
	COMP. MODEL	COMP. S/N	PUBL. DATE	01/08/2008
	RECEIVING DATE	FINAL INSPEC	REVISION NO.	00
	DELIVERY DATE	WO NO.	PAGE	1 of 1

BELOW ARE THE ITEMS RELATED TO DELIVERY INSPECTION AND WHILE YOU DOING THIS INSPECTION JOB, PLEASE REFERRING TO THE REMANUFACTURED COMPONENTS - STANDARD CONFIGURATION.

✓ : GOOD CONDITION X : BAD CONDITION ⊗ : NONE

SKETCH DRAWING OR PHOTOES	NO.	ITEM	INSPECTION CONDITION	CHECK	
				IN	OUT
 		Comp. Model			
		Aplication			
		Serial Number			
		Wo No.			
	1	Sun Gear (19 teeth)			
	2	Planetary gear (31 teeth)			
	3	Planetary gear Shaft			
	4	Ring Gear (87 teeth)			
	5	Inner hub			
	6	Wheel Hub			
	7	Drive shaft			
	8	Reservoir tank			
	9	Cylinder			
	10	Gear			
	11	Stud bolt			
	12	Cover and monting bolt			
		Stand			
		Painting			
		Packing			

NOTE : PLEASE MARK " ✓ " FOLLOWING TO THE ACTUAL CONDITION

OIL LEVEL	EMPTY OR DRAINED	
	PROPER LEVEL	

REMARKS :

Jakarta,

Inspected by,

Aproved by,

QA.Inspector

SPV/GL



PT SAPTAINDRA SEJATI

FINAL DRIVE RH

DISASSEMBLY

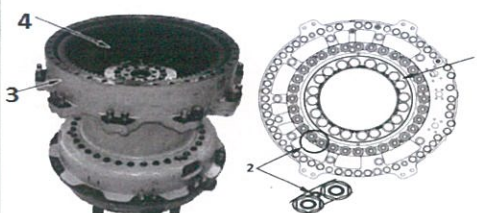
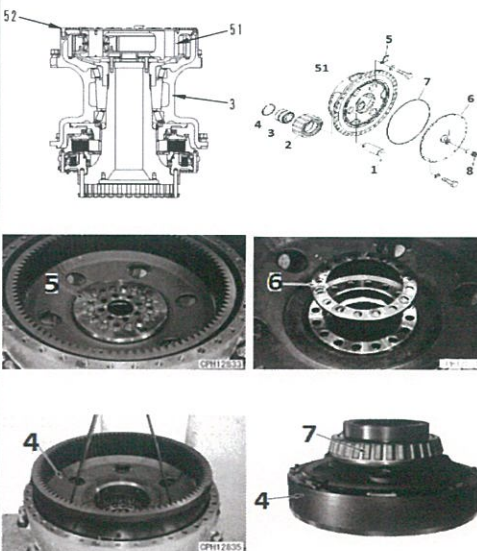
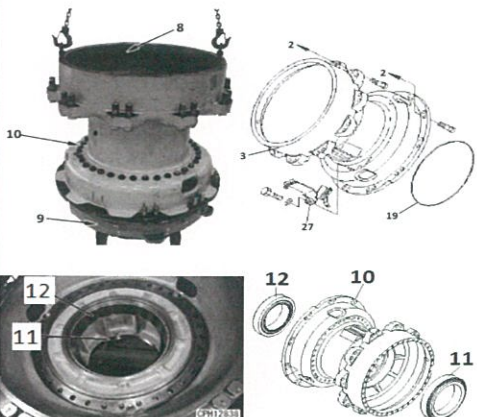
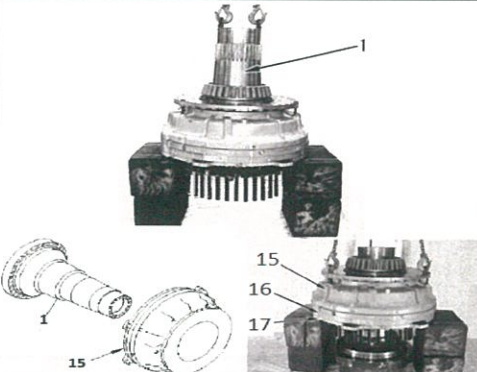
CHECKSHEET

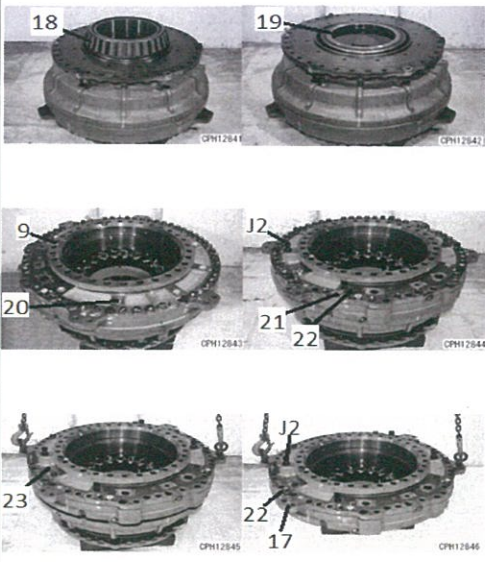
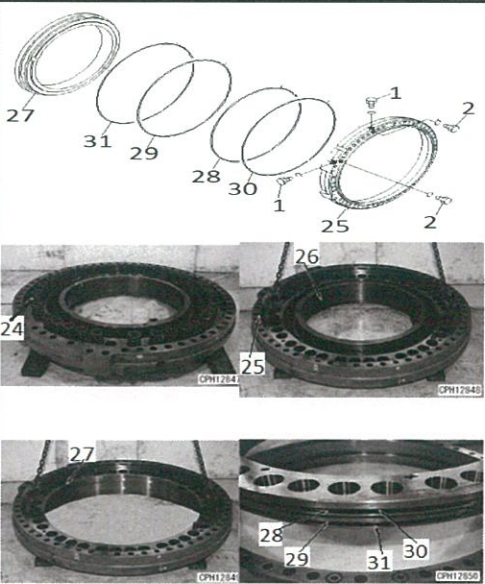
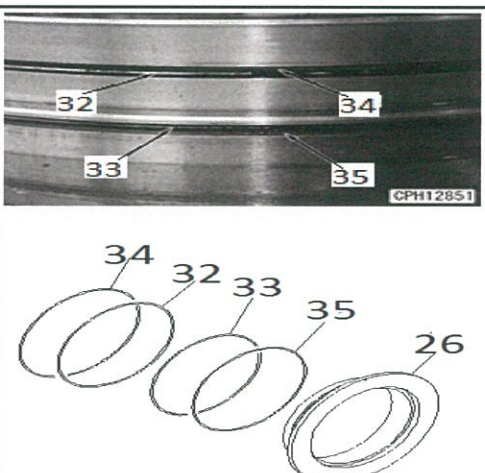
MACHINE MODEL

HD785-7

RO NO :

POWERTRAIN SECTION

DISASSEMBLY CHECK SHEET			WO NUMBER		DOCUMENT NO		RC/QR02/PT/04/01			
FINAL DRIVE			COMP.MODEL		PUBLIC DATE					
HD 785 - 7			COMP.S/N		REVISION NO					
			UNIT MODEL		PAGE					
PROCESS		START			DISASSEMBLY BY					
		FINISH								
NO	PART TO REMOVE OR INSPECT	CONDITIONS TOBE INSPECTION	MECH	GL	SKECTH DRAWING OR PHOTOES		REMARK			
1	SETTING OF REAR AXLE ASSEMBLY	⚠ IMPORTANT ⚠ Set the rear axle assembly vertically on stand and check worn abnormally, crack, chipping, scrath, scoring and threet of component. Rear axle assembly: 3,250 kg - Remove 4 mounting bolts (2) of shaft (1). - Set rear axle assembly (3) with planetary carrier (4)								
2	PLANETARY CARRIER ASSEMBLY AND RING GEAR	⚠ IMPORTANT ⚠ Set the rear axle assembly vertically and check worn abnormally, crack, chipping, scrath, scoring and threet of component. - Remove bolt (52) and Remove planetary carrier assembly (51) - Remove bracket (5) And remove gear (2) bearing (3) snap ring (4) ball and shaft (1) from carrier (51) with pressfit method - Sling ring gear temporarily and remove retainer (5) - Remove shims (6) - a check the quatinty and thickness of the shims <table border="1"><tr><td>Actual</td></tr><tr><td> </td></tr></table> - Lift off ring gear (4) and Remove bearing (7) from ring gear (4) with tracking methode	Actual							
Actual										
3	REAR AXLE CASE	⚠ IMPORTANT ⚠ a Make a match mark on rear axle case (8) and brake assembly (9). and check worn abnormally, crack, chipping, scrath, scoring and threet of component. - Remove rear axle case mounting bolts (10) and Sling rear axle case (8). - Remove bracket (27) and studs (2) from housing (8) - Sling rear axle and brake assembly (8) and disconnect it from the shaft. k Sling the assembly so carefully that the shaft will not overturn. - Remove the outer races (11), (12) from the rear axle. With pressfit method								
4	SHAFT	⚠ IMPORTANT ⚠ Set the rear axle assembly vertically and check worn abnormally, crack, chipping, scrath, scoring and threet of component. Shaft : 750 Kg & brake assembly : 1150 Kg - Lift rear brake and shaft assembly (1),and reposition it at a height of approx. 400 mm - Lift down shaft (1) to disconnect. - Remove brake assembly. - a Put counter marks on service brake assembly (15), parking brake cylinder (16) and parking brake cover (17).								

DISASSEMBLY CHECK SHEET		WO NUMBER	DOCUMENT NO	RC/QR02/PT/04/01		
FINAL DRIVE		COMP.MODEL	PUBLIC DATE			
HD 785 - 7		COMP.S/N	REVISION NO			
		UNIT MODEL	PAGE			
PROCESS		START	DISASSEMBLY BY			
		FINISH				
NO	PART TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTION	MECH	GL	SKETCH DRAWING OR PHOTOS	REMARK
5	PARKING BRAKE HOUSING	⚠ IMPORTANT ⚠ Set the rear axle assembly vertically and check worn abnormally, crack, chipping, scratch, scoring and threat of component. Bearing : 50 Kg & Parking brake assembly : 495 Kg - Remove bearing(18)Remove floating seal(19). Reverse and place brake assembly (9). - Remove plug (20). Install parking brake manual releasing bolts (J2) at least 4 locations in even distribution.Tighten nuts evenly to retract parking brake spring.a Tightening length: Approx. 10 mm - Remove 45 parking brake mounting bolts - Do not remove parking brake cover mounting bolts (22) having smaller washer diameter. - Lift and remove parking brake assembly (23) - Remove parking brake cover bolts (22). Loosen nuts of parking brake manual a releasing bolts (J2) evenly, and remove The piston and cylinder may be scratched, if they are not loosened evenly. - Remove parking brake cover (17).				
6	PARKING BRAKE PISTON AND PARKING BRAKE CYLINDER ASSEMBLY	⚠ IMPORTANT ⚠ Set the rear axle assembly vertically and check worn abnormally, crack, chipping, scratch, scoring, Bending of spring (24) and threat of component. - Remove parking brake spring (24). - Remove parking brake piston and parking brake cylinder assembly (25) from service brake piston (26). Parking brake piston and parking brake cylinder assembly: 310 kg Service brake piston: 80 kg - Remove parking brake piston (27). - Parking brake piston: 95 kg - Remove seal rings (28) and (29), and backup rings (30) and (31) from parking brake piston.				
7	SERVICE BRAKE PISTON ASSEMBLY	⚠ IMPORTANT ⚠ Set the rear axle assembly vertically and check worn abnormally, crack, chipping, scratch, scoring, Bending of spring (24) and threat of component. Remove seal rings (32) and (33), and backup rings (34) and (35) from service brake piston (26)				

DISASSEMBLY CHECK SHEET FINAL DRIVE HD 785 - 7	WO NUMBER		DOCUMENT NO	RC/QR02/PT/04/01
	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	
	UNIT MODEL		PAGE	

PROCESS	START		DISASSEMBLY BY		
	FINISH				

NO	PART TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTED	MECH	GL	SKETCH DRAWING OR PHOTOS	REMARK
8	DISC AND PLATE	⚠ IMPORTANT ⚠ a Make a match mark on rear axle case (8) and brake assembly (9). and check worn abnormally, crack, chipping, scratch, scoring and threat of component. - Remove lock plates (36). - Remove cork-attached plate (37). - Remove disc (38). - Remove plate (39). a 9 discs and 8 plates are assembled. - Remove plate having facing (40). a The facing is affixed to the space side. - Remove spacer (41). Spacer: 20 kg - Remove mounting bolts (42) and remove inner gear (43). Inner gear: 60 kg				
9	RETAINER AND GEAR ASSEMBLY	⚠ IMPORTANT ⚠ a Make a match mark on rear axle case (8) and brake assembly (9). and check worn abnormally, crack, chipping, scratch, scoring and threat of component. - Lift outer gear (44) temporarily, and remove floating seal lock bolt (J1). - Remove outer gear (44). Outer gear: 200 kg - Reverse outer gear (44) and remove floating seal (45). - Remove floating seal (47) from cage (46). Reverse cage. Cage: 130 kg - Remove floating seal (48) from cage (46).				

NOTE :

INSPECTION BY

MECHANIC

APPROVED BY

GL /QA



PT SAPTAINDRA SEJATI

FINAL DRIVE RH

MEASUREMENT

CHECKSHEET

MACHINE MODEL

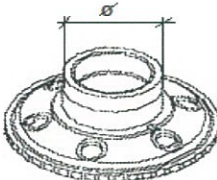
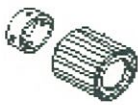
HD785-7

RO NO :

POWERTRAIN SECTION

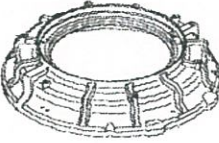
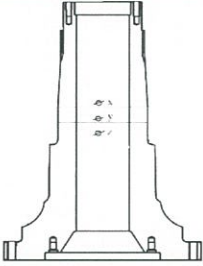
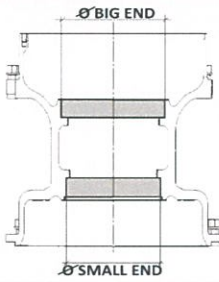
VISUAL & MEASUREMENT CHECK SHEET FINAL DRIVE HD 785 - 7		WO NUMBER	DOCUMENT NO	RC/QR03/PT/04/01
		COMP.MODEL	PUBLIC DATE	
		COMP.S/N	REVISION NO	
		UNIT MODEL	PAGE	

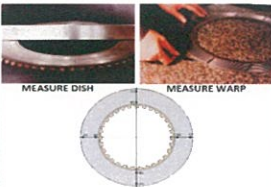
PROCESS	START	FINISH	MEASUREMENT BY		
---------	-------	--------	----------------	--	--

NO	PART TO BE INSPECTED	CHECKING POINT	SKECTH DRAWING OR PHOTOES	VISUAL CHECK & MEASUREMENT																																	
1	CARRIER HOUSING (P/N 561-22-72520)	- Planetary shaft hole - O - Ring Groove - All round		<table border="1" style="font-size: 8px;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Scratch</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Wear</td><td></td></tr> </table>	Contens check	Actual	Normal		Crack		Scratch		Broken		Wear		<table border="1" style="font-size: 8px;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>			U/A	U/R	R/N															
Contens check	Actual																																				
Normal																																					
Crack																																					
Scratch																																					
Broken																																					
Wear																																					
U/A	U/R	R/N																																			
2	CARRIER GEAR (P/N 561-22-72531)	- Gear teeth - Bearing fitting portion - Snap ring groove		<table border="1" style="font-size: 8px;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Scratch</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Wear</td><td></td></tr> </table>	Contens check	Actual	Normal		Crack		Scratch		Broken		Wear		<table border="1" style="font-size: 8px;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>			U/A	U/R	R/N															
Contens check	Actual																																				
Normal																																					
Crack																																					
Scratch																																					
Broken																																					
Wear																																					
U/A	U/R	R/N																																			
3	CARRIER SHAFT (P/N 561-22-72511)	- Bearing fitting portion - Carrier bore fitting portion - Ball fitting portion		<table border="1" style="font-size: 8px;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Scratch</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Wear</td><td></td></tr> </table>	Contens check	Actual	Normal		Crack		Scratch		Broken		Wear		<table border="1" style="font-size: 8px;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>			U/A	U/R	R/N															
Contens check	Actual																																				
Normal																																					
Crack																																					
Scratch																																					
Broken																																					
Wear																																					
U/A	U/R	R/N																																			
4	HUB (P/N 561-22-72313)	- Internal spline teeth - External gear teeth - Bearing fitting portion - Retainer fitting portion		<table border="1" style="font-size: 8px;"> <tr><th>X</th><th>Y</th><th>Z</th></tr> <tr><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td></tr> </table> Remark :	X	Y	Z										<table border="1" style="font-size: 8px;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Scratch</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Wear</td><td></td></tr> </table>	Contens check	Actual	Normal		Crack		Scratch		Broken		Wear		<table border="1" style="font-size: 8px;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>		U/A	U/R	R/N			
X	Y	Z																																			
Contens check	Actual																																				
Normal																																					
Crack																																					
Scratch																																					
Broken																																					
Wear																																					
U/A	U/R	R/N																																			
5	SPACER & SUN GEAR SPACER (P/N 568-22-12442) GEAR (P/N 561-22-72420)	- Gear teeth - Internal spline teeth - Side surface		<table border="1" style="font-size: 8px;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Scratch</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Wear</td><td></td></tr> </table>	Contens check	Actual	Normal		Crack		Scratch		Broken		Wear		<table border="1" style="font-size: 8px;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>			U/A	U/R	R/N															
Contens check	Actual																																				
Normal																																					
Crack																																					
Scratch																																					
Broken																																					
Wear																																					
U/A	U/R	R/N																																			
6	RETAINER (P/N 561-22-72210)	- Matting Face - All round		<table border="1" style="font-size: 8px;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Scratch</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Wear</td><td></td></tr> </table>	Contens check	Actual	Normal		Crack		Scratch		Broken		Wear		<table border="1" style="font-size: 8px;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>			U/A	U/R	R/N															
Contens check	Actual																																				
Normal																																					
Crack																																					
Scratch																																					
Broken																																					
Wear																																					
U/A	U/R	R/N																																			
7	CYLINDER LH (P/N 561-34-71114) RH(P/N 561-34-75112)	- Friction surface - Seal ring contact surface - O-ring groove - All round		<table border="1" style="font-size: 8px;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Scratch</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Wear</td><td></td></tr> </table>	Contens check	Actual	Normal		Crack		Scratch		Broken		Wear		<table border="1" style="font-size: 8px;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>			U/A	U/R	R/N															
Contens check	Actual																																				
Normal																																					
Crack																																					
Scratch																																					
Broken																																					
Wear																																					
U/A	U/R	R/N																																			
8	CYLINDER LH (PN 561-33-71115) RH (PN 561-33-75111)	- Friction surface - Seal ring contact surface - O-ring groove - All round		<table border="1" style="font-size: 8px;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Scratch</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Wear</td><td></td></tr> </table>	Contens check	Actual	Normal		Crack		Scratch		Broken		Wear		<table border="1" style="font-size: 8px;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>			U/A	U/R	R/N															
Contens check	Actual																																				
Normal																																					
Crack																																					
Scratch																																					
Broken																																					
Wear																																					
U/A	U/R	R/N																																			
9	PISTON (P/N 561-33-71123)	- Friction surface - seal ring contact surface - seal ring groove - All round		<table border="1" style="font-size: 8px;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Scratch</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Wear</td><td></td></tr> </table>	Contens check	Actual	Normal		Crack		Scratch		Broken		Wear		<table border="1" style="font-size: 8px;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>			U/A	U/R	R/N															
Contens check	Actual																																				
Normal																																					
Crack																																					
Scratch																																					
Broken																																					
Wear																																					
U/A	U/R	R/N																																			

VISUAL & MEASUREMENT CHECK SHEET FINAL DRIVE HD 785 - 7			WO NUMBER	DOCUMENT NO	RC/QR03/PT/04/01
			COMP.MODEL	PUBLIC DATE	
			COMP.S/N	REVISION NO	
			UNIT MODEL	PAGE	

PROCESS	START FINISH	MEASUREMENT BY	
---------	-----------------	----------------	--

NO	PART TO BE INSPECTED	CHECK POINT	SKECTH DRAWING OR PHOTOS	VISUAL CHECK & MEASUREMENT																																																		
10	PISTON (P/N 561-33-71123)	- Friction surface - Seal ring contact surface - Seal ring groove - All round		<table border="1" style="display: inline-table; vertical-align: top;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Scratch</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Wear</td><td></td></tr> </table> <table border="1" style="display: inline-table; vertical-align: top; margin-left: 10px;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>	Contens check	Actual	Normal		Crack		Scratch		Broken		Wear		U/A	U/R	R/N																																			
Contens check	Actual																																																					
Normal																																																						
Crack																																																						
Scratch																																																						
Broken																																																						
Wear																																																						
U/A	U/R	R/N																																																				
11	RETAINER (P/N 561-88-71171)	- Oil seal contact surface - Floating seal contact surface - O-ring groove		<table border="1" style="display: inline-table; vertical-align: top;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Scratch</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Wear</td><td></td></tr> </table> <table border="1" style="display: inline-table; vertical-align: top; margin-left: 10px;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>	Contens check	Actual	Normal		Crack		Scratch		Broken		Wear		U/A	U/R	R/N																																			
Contens check	Actual																																																					
Normal																																																						
Crack																																																						
Scratch																																																						
Broken																																																						
Wear																																																						
U/A	U/R	R/N																																																				
12	RETAINER (P/N 561-33-71931)	- Floating seal contact surface - O-ring groove - All round		<table border="1" style="display: inline-table; vertical-align: top;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Scratch</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Wear</td><td></td></tr> </table> <table border="1" style="display: inline-table; vertical-align: top; margin-left: 10px;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>	Contens check	Actual	Normal		Crack		Scratch		Broken		Wear		U/A	U/R	R/N																																			
Contens check	Actual																																																					
Normal																																																						
Crack																																																						
Scratch																																																						
Broken																																																						
Wear																																																						
U/A	U/R	R/N																																																				
13	SPRING SMALL (P/N 561-34-71210) LARGE (P/N 561-34-71220)	- Contact surface - All round		<table border="1" style="display: inline-table; vertical-align: top;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Fracture</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Bending</td><td></td></tr> </table> <table border="1" style="display: inline-table; vertical-align: top; margin-left: 10px;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table> Remark :	Contens check	Actual	Normal		Crack		Fracture		Broken		Bending		U/A	U/R	R/N																																			
Contens check	Actual																																																					
Normal																																																						
Crack																																																						
Fracture																																																						
Broken																																																						
Bending																																																						
U/A	U/R	R/N																																																				
14	SENSOR (P/N 566-88-66141)	All round		<table border="1" style="display: inline-table; vertical-align: top;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Fructure</td><td></td></tr> <tr><td></td><td></td></tr> <tr><td></td><td></td></tr> </table> <table border="1" style="display: inline-table; vertical-align: top; margin-left: 10px;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>	Contens check	Actual	Normal		Broken		Fructure						U/A	U/R	R/N																																			
Contens check	Actual																																																					
Normal																																																						
Broken																																																						
Fructure																																																						
U/A	U/R	R/N																																																				
15	GEAR OUTER LH (P/N 561-88-71215) RH (P/N 561-88-75212)	- Floating seal contact surface - Oil seal contact surface - Internal spline teeth - All round		<table border="1" style="display: inline-table; vertical-align: top;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Scratch</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Wear</td><td></td></tr> </table> <table border="1" style="display: inline-table; vertical-align: top; margin-left: 10px;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>	Contens check	Actual	Normal		Crack		Scratch		Broken		Wear		U/A	U/R	R/N																																			
Contens check	Actual																																																					
Normal																																																						
Crack																																																						
Scratch																																																						
Broken																																																						
Wear																																																						
U/A	U/R	R/N																																																				
16	TUBE (P/N 561-22-73212)	- Bearing fitting portion - Spline teeth & O-ring groove - All round Shaft <table border="1" style="display: inline-table; vertical-align: top;"> <tr><th>Standart</th><th>tolerance</th></tr> <tr><td>320</td><td>-0,034</td></tr> </table>	Standart	tolerance	320	-0,034		<table border="1" style="display: inline-table; vertical-align: top;"> <tr><th>X</th><th>Y</th><th>Z</th></tr> <tr><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td></tr> </table> Remark : <table border="1" style="display: inline-table; vertical-align: top; margin-left: 10px;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Scratch</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Wear</td><td></td></tr> </table> <table border="1" style="display: inline-table; vertical-align: top; margin-left: 10px;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>	X	Y	Z										Contens check	Actual	Normal		Crack		Scratch		Broken		Wear		U/A	U/R	R/N																			
Standart	tolerance																																																					
320	-0,034																																																					
X	Y	Z																																																				
Contens check	Actual																																																					
Normal																																																						
Crack																																																						
Scratch																																																						
Broken																																																						
Wear																																																						
U/A	U/R	R/N																																																				
17	HUB (P/N 561-88-72710)	- Bearing fitting portion - thread portion & Matting Face Big end <table border="1" style="display: inline-table; vertical-align: top;"> <tr><th>Standart</th><th>tolerance</th></tr> <tr><td>523.875</td><td>-0,063</td></tr> </table> Small end <table border="1" style="display: inline-table; vertical-align: top;"> <tr><th>Standart</th><th>tolerance</th></tr> <tr><td>480</td><td>-0,045</td></tr> </table>	Standart	tolerance	523.875	-0,063	Standart	tolerance	480	-0,045		Big end <table border="1" style="display: inline-table; vertical-align: top;"> <tr><th>X</th><th>Y</th><th>Z</th></tr> <tr><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td></tr> </table> Small end <table border="1" style="display: inline-table; vertical-align: top;"> <tr><th>X</th><th>Y</th><th>Z</th></tr> <tr><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td></tr> </table> <table border="1" style="display: inline-table; vertical-align: top; margin-left: 10px;"> <tr><th>Contens check</th><th>Actual</th></tr> <tr><td>Normal</td><td></td></tr> <tr><td>Crack</td><td></td></tr> <tr><td>Scratch</td><td></td></tr> <tr><td>Broken</td><td></td></tr> <tr><td>Wear</td><td></td></tr> </table> <table border="1" style="display: inline-table; vertical-align: top; margin-left: 10px;"> <tr><th>U/A</th><th>U/R</th><th>R/N</th></tr> <tr><td></td><td></td><td></td></tr> </table>	X	Y	Z										X	Y	Z										Contens check	Actual	Normal		Crack		Scratch		Broken		Wear		U/A	U/R	R/N			
Standart	tolerance																																																					
523.875	-0,063																																																					
Standart	tolerance																																																					
480	-0,045																																																					
X	Y	Z																																																				
X	Y	Z																																																				
Contens check	Actual																																																					
Normal																																																						
Crack																																																						
Scratch																																																						
Broken																																																						
Wear																																																						
U/A	U/R	R/N																																																				

VISUAL & MEASUREMENT CHECK SHEET FINAL DRIVE HD 785 - 7			WO NUMBER		DOCUMENT NO		RC/QR03/PT/04/01																	
			COMP.MODEL		PUBLIC DATE																			
			COMP.S/N		REVISION NO																			
			UNIT MODEL		PAGE																			
PROCESS		START			MEASUREMENT BY																			
		FINISH																						
NO	PART TO BE INSPECTED	CHECK POINT	SKEETH DRAWING OR PHOTOOES		VISUAL CHECK & MEASUREMENT																			
18	PLATE (P/N 561-33-71230)	- Deformation - tooth flank wear on external gear - Heat spot or metal transfer <table border="1"> <tr><th colspan="2">THICKNESS</th></tr> <tr> <th>Standart</th><th>tolerance</th></tr> <tr> <td>2,40</td><td>-0,25</td></tr> </table> <table border="1"> <tr><th colspan="2">DISH & WRAP</th></tr> <tr> <th>Standart</th><th>tolerance</th></tr> <tr> <td>0,5</td><td>0,2</td></tr> </table>	THICKNESS		Standart	tolerance	2,40	-0,25	DISH & WRAP		Standart	tolerance	0,5	0,2			THICKNESS							
			THICKNESS																					
			Standart	tolerance																				
			2,40	-0,25																				
			DISH & WRAP																					
			Standart	tolerance																				
			0,5	0,2																				
			1	2	3	4	5	6	7	8														
DISH & WRAP																								
1	2	3	4	5	6	7	8																	
Remark																								
						U/A	U/R	R/N																

[illegible]



PT SAPTAINDRA SEJATI

FINAL DRIVE RH

ASSEMBLY

CHECKSHEET

MACHINE MODEL

HD785-7

RO NO :

POWERTRAIN SECTION

ASSEMBLY CHECK SHEET

FINAL DRIVE RH

HD 785 - 7

WO NUMBER

COMP.MODEL

COMP.S/N

UNIT MODEL

DOCUMENT NO

PUBLIC DATE

REVISION NO

PAGE

R 01

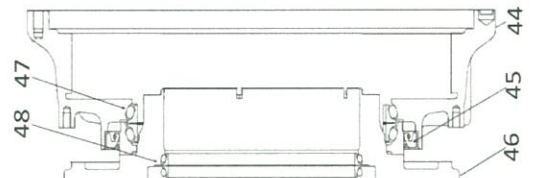
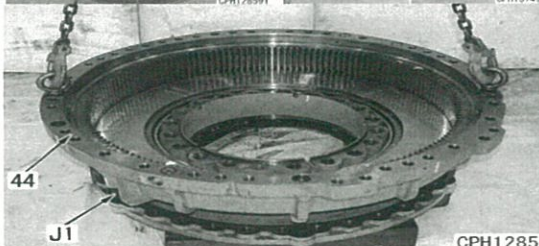
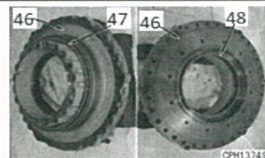
6 of 9

PROCESS

START

FINISH

ASSEMBLY BY

NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH	GL/QA	SKETCH DRAWING OR PHOTHOES																
1	CAGE AND OUTER GEAR ASSEMBLY	⚠ IMPORTANT ⚠ Inspect each part thoroughly and confirm that the part is free from dirt/dust, scratches. Thoroughly dry and clean conditions of the floating seal and O-ring contact surface of the housing. apply force evenly on the O-ring of the floating seal for tightening so as not to let it twist. - Install floating seals (47) and (48) to cage (46). - Install floating seal (45) to outer gear (44). - Install outer gear (44) to cage. - Outer gear: 200 kg - Fix outer gear and cage with floating seal lock bolt (J1). a Tighten until the inner gear assembly contacts to the outer gear assembly tightly. Tightening amount: Approx. 3 mm a After assembling the rear brake assembly to the machine, remove bolt (J1).																			
2	DISC AND PLATE ASSEMBLY	⚠ IMPORTANT ⚠ Inspect each part thoroughly and confirm that the part is free from dirt/dust, scratches.  Engine oil SAE 30 - Install inner gear (43). - Inner gear: 60 kg - Install bolt (42)  Tightening bolt (42) <table><tr><th>Standard</th><th>Actual</th></tr><tr><td>175 ± 20 kgm</td><td></td></tr></table> - Install spacer (41). - Spacer: 20 kg - Install plate having facing (40). - a Facing-side toward the spacer. - P/N 561-33-71261 Thickness of Plate having face (Damper) 3.40mm - Install disc (38). P/N 561-33-71240 Thickness of disc 6.50 mm <table><tr><th>Standard</th><th>Actual</th></tr><tr><td>9 pcs</td><td></td></tr></table> - Install plate (39). P/N 561-33-71230 Thickness of plate 2.40 mm <table><tr><th>Standard</th><th>Actual</th></tr><tr><td>8 pcs</td><td></td></tr></table> - Install cork-attached plate (37). - a Cork-side toward the piston. - P/N 561-33-71271 Thickness of demper 6.90 mm - Install lock plate (36).  Tighting bolt (36) <table><tr><th>Standard</th><th>Actual</th></tr><tr><td>3.5 kgm</td><td></td></tr></table>	Standard	Actual	175 ± 20 kgm		Standard	Actual	9 pcs		Standard	Actual	8 pcs		Standard	Actual	3.5 kgm				<
Standard	Actual																				
175 ± 20 kgm																					
Standard	Actual																				
9 pcs																					
Standard	Actual																				
8 pcs																					
Standard	Actual																				
3.5 kgm																					

RANDOM ITEM CHECK

Name :

Name :

QA INSPECTOR

PROD. SUPERVISOR

ASSEMBLY CHECK SHEET
FINAL DRIVE RH

HD 785 - 7

WO NUMBER

COMP.MODEL

COMP.S/N

UNIT MODEL

DOCUMENT NO

PUBLIC DATE

REVISION NO

PAGE

R 01

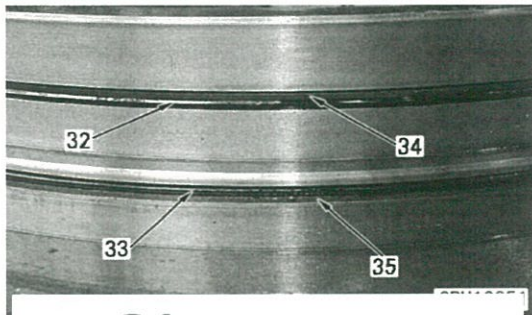
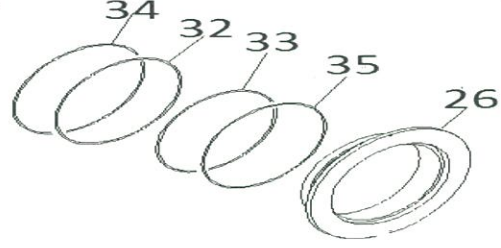
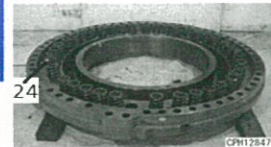
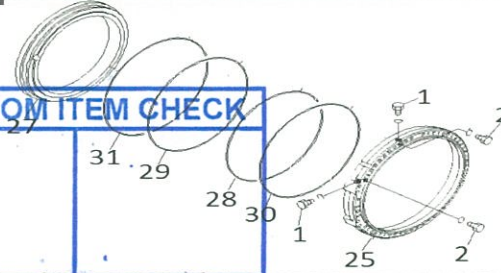
7 of 9

PROCESS

START

FINISH

ASSEMBLY BY

NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH	GL/QA	SKETCH DRAWING OR PHOTHOES																							
3	SERVICE BRAKE ASSEMBLY	⚠ IMPORTANT ⚠ Inspect each part thoroughly and confirm that the part is free from dirt/dust, scratches.  Engine oil SAE 30 - Install seal rings (32) and (33), and backup rings (34) and (35) to service brake piston (26). - Install seal ring to the pressure receiving side (inside). - Install service brake piston (26).			 																							
4	PARKING BRAKE ASSEMBLY	⚠ IMPORTANT ⚠ Inspect each part thoroughly and confirm that the part is free from dirt/dust, scratches.  Engine oil SAE 30 - Install seal rings (28) and (29), and backup rings (30) and (31) to piston. - Install seal ring to the pressure receiving side (inside). ⚠ IMPORTANT ⚠ - Fit O-ring and Install plug (2) 2 pcs P/N plug (2) 07040-11409  Tightening plug (2) <table><tr><th>Standard</th><th>Actual</th></tr><tr><td>5.0 ± 0.5 Kgm</td><td></td></tr></table> - Install piston (27) to parking brake cylinder - If the cylinder and piston are not aligned, their sliding surfaces may be scratched. - Install the set of large and small springs (24) - Large spring P/N 561-34-71220<table><tr><th>Standard</th><th>Actual</th></tr><tr><td>39 pcs</td><td></td></tr></table> - Small spring P/N 561-34-71210<table><tr><th>Standard</th><th>Actual</th></tr><tr><td>39 pcs</td><td></td></tr></table> - Install parking brake cover (23). - Install parking brake manual releasing bolts (J2), nuts and washers at least 4 locations in even distribution. Tighten nuts evenly to retract parking brake spring. a Tightening length: Approx. 10 mm - Install cover mounting bolts (22).  Tightening bolt (36)<table><tr><th>Standard</th><th>Actual</th></tr><tr><td>56.0 ± 6.0 Kgm</td><td></td></tr></table> - Install parking brake assembly (23) to service brake assembly (15). - Install 45 parking brake mounting bolts.  Tightening bolt<table><tr><th>Standard</th><th>Actual</th></tr><tr><td>56.0 ± 6.0 Kgm</td><td></td></tr></table> - Remove release bolt (J2). - Install plug (20)  Tightening bolt (20)<table><tr><th>Standard</th><th>Actual</th></tr><tr><td>5.0 ± 0.5 Kgm</td><td></td></tr></table>	Standard	Actual	5.0 ± 0.5 Kgm		Standard	Actual	39 pcs		Standard	Actual	39 pcs		Standard	Actual	56.0 ± 6.0 Kgm		Standard	Actual	56.0 ± 6.0 Kgm		Standard	Actual	5.0 ± 0.5 Kgm			RANDOM ITEM CHECK Name : _____ CA INSPECTOR / PROD. SUPERVISOR 
Standard	Actual																											
5.0 ± 0.5 Kgm																												
Standard	Actual																											
39 pcs																												
Standard	Actual																											
39 pcs																												
Standard	Actual																											
56.0 ± 6.0 Kgm																												
Standard	Actual																											
56.0 ± 6.0 Kgm																												
Standard	Actual																											
5.0 ± 0.5 Kgm																												

ASSEMBLY CHECK SHEET FINAL DRIVE RH

HD 785 - 7

WO NUMBER

COMP.MODEL

COMP.S/N

UNIT MODEL

DOCUMENT NO

PUBLIC DATE

REVISION NO

PAGE

R 01

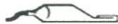
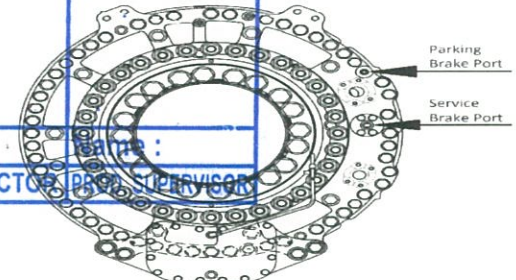
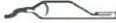
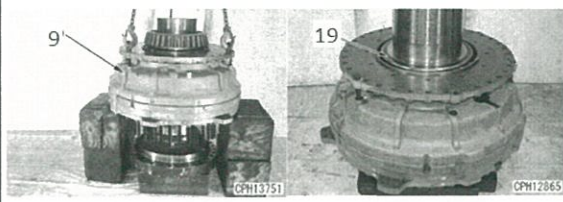
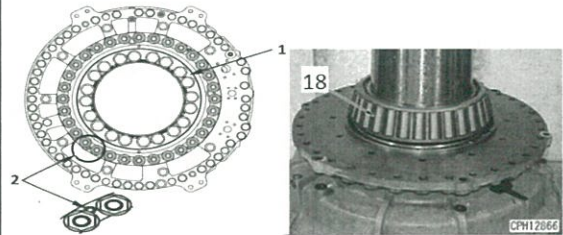
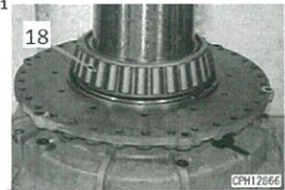
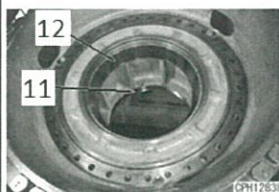
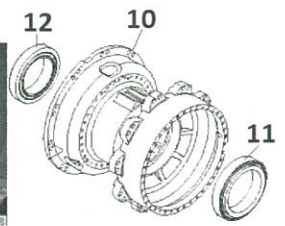
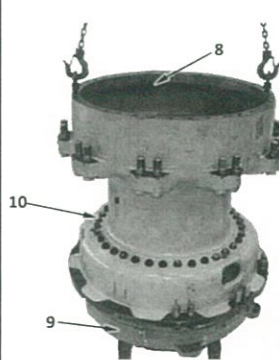
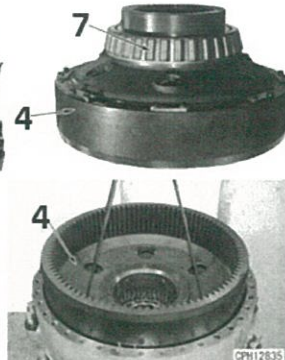
8 of 9

PROCESS

START

FINISH

ASSEMBLY BY

NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH	GL/QA	SKETCH DRAWING OR PHOTHOES												
5	TEST LEAK PARKING BRAKE AND SERVICE BRAKE	⚠ IMPORTANT ⚠ Inspect each part thoroughly and confirm that the part is free from dirt/dust, scratches.  Engine oil SAE 10 - Check leak on piston brake and parking brake assembly - check with oil pressure $100 \pm 7 \text{ kg/cm}^2$ at min 15 secon Parking brake test leak <table><tr><th>Standard</th><th>Actual</th></tr><tr><td>$100 \pm 7 \text{ kg/cm}^2$</td><td></td></tr><tr><td>15 secon</td><td></td></tr></table> Service brake test leak <table><tr><th>Standard</th><th>Actual</th></tr><tr><td>$100 \pm 7 \text{ kg/cm}^2$</td><td></td></tr><tr><td>15 secon</td><td></td></tr></table>	Standard	Actual	$100 \pm 7 \text{ kg/cm}^2$		15 secon		Standard	Actual	$100 \pm 7 \text{ kg/cm}^2$		15 secon				 RANDOM ITEM CHECK Name : _____ Date : _____ QA INSPECTOR : _____ 
Standard	Actual																
$100 \pm 7 \text{ kg/cm}^2$																	
15 secon																	
Standard	Actual																
$100 \pm 7 \text{ kg/cm}^2$																	
15 secon																	
6	SHAFT AND BRAKE ASSEMBLY	⚠ IMPORTANT ⚠ Inspect each part thoroughly and confirm that the part is free from dirt/dust, scratches.  Engine oil SAE 30 - Place shaft on stand - Place shaft (1) with the flange face down. Shaft: 750 kg - Sling the brake assembly (9) down. - Brake assembly: 1,100 kg - install 4 shaft mounting bolts (2).  Tightening bolt (2) <table><tr><th>Standard</th><th>Actual</th></tr><tr><td>$11.5 \pm 0.5 \text{ Kgm}$</td><td></td></tr></table> - Install floating seal (19). - Install bearing (18). - Bearing: 18 kg	Standard	Actual	$11.5 \pm 0.5 \text{ Kgm}$				  								
Standard	Actual																
$11.5 \pm 0.5 \text{ Kgm}$																	
7	HUB AND RING GEAR ASSEMBLY	⚠ IMPORTANT ⚠ Inspect each part thoroughly and confirm that the part is free from dirt/dust, scratches. Be careful not to apply impact or drop the parts during installation. Even the slightest damage on the floating seal can cause oil leaks.  Engine oil SAE 30 - Install outer race (12). And install outer race (11). - Reverse rear axle case (8) and install to brake assembly (9). - Rear axle case: 1,000 kg  Tightening bolt (10) <table><tr><th>Standard</th><th>Actual</th></tr><tr><td>$56.0 \pm 6.0 \text{ Kgm}$</td><td></td></tr></table> - Install bearing (7) to ring gear (4). - Bearing: 50 kg - Install ring gear assembly (4) to rear axle case (8) - Ring gear and bearing assembly: 350 kg	Standard	Actual	$56.0 \pm 6.0 \text{ Kgm}$				   								
Standard	Actual																
$56.0 \pm 6.0 \text{ Kgm}$																	

ASSEMBLY CHECK SHEET

FINAL DRIVE RH

HD 785 - 7

WO NUMBER

COMP.MODEL

COMP.S/N

UNIT MODEL

DOCUMENT NO

PUBLIC DATE

REVISION NO

PAGE

R 01

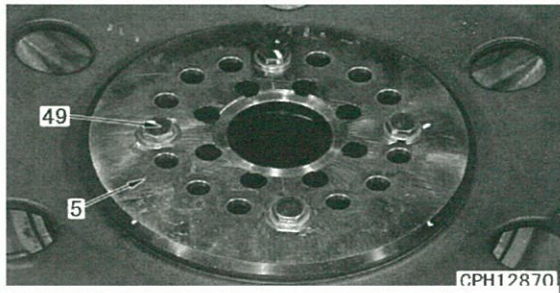
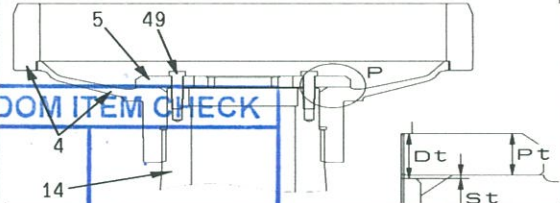
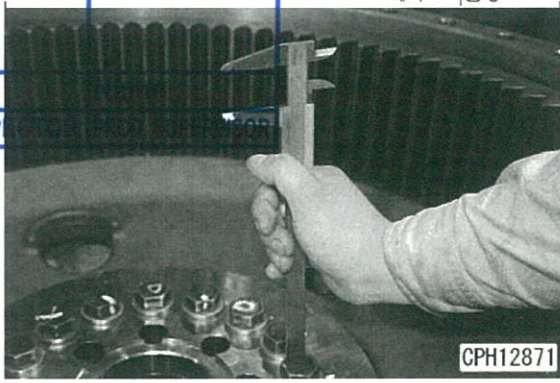
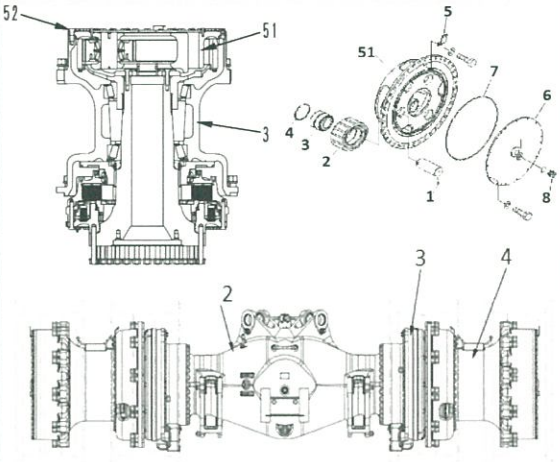
9 of 9

PROCESS

START

FINISH

ASSEMBLY BY

NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH	GL/QA	SKETCH DRAWING OR PHOTHOES																						
8	ADJUSTMENT PRELOAD OF BEARING ASSEMBLY	⚠ IMPORTANT ⚠ Inspect each part thoroughly and confirm that the part is free from dirt/dust, scratches. Thoroughly dry and clean conditions of the shims and shaft contact surface . - Measure thickness (Pt) of retainer (5). Install retainer (5) without shim, and use 4 bolts (49) Tightening bolt (49) for adjustment <table border="1"> <thead> <tr> <th>Standard</th> <th>Actual</th> </tr> </thead> <tbody> <tr> <td>5.9 - 6.9 Kgm</td> <td></td> </tr> </tbody> </table> - Measure distance (depth Dt) from spacer end surface to shaft end surface. - Measure thickness of the retainer by 3 positions, and obtain the average (Pt). Adjust the number of shims so that it conforms the equation below, then install shims (6). Shim thickness (St) = (Dt) - (Pt) + 0.3 mm Thickness of shims <table border="1"> <thead> <tr> <th>Actual</th> </tr> </thead> <tbody> <tr> <td></td> </tr> </tbody> </table> - Install retainer (5), and tighten mounting Tightening bolt (49) <table border="1"> <thead> <tr> <th>Standard</th> <th>Actual</th> </tr> </thead> <tbody> <tr> <td>56.0 ± 6.0 Kgm</td> <td></td> </tr> </tbody> </table> - Install snap ring (4) and bearing (3) to gear (2) - Install gear (2) to carrier housing (51) - Install sahft and ball (1) to carrier housing (51) a with pressfit methode Engine oil SAE 30 - Install carrier assembly (51) Tightening bolt (52) <table border="1"> <thead> <tr> <th>Standard</th> <th>Actual</th> </tr> </thead> <tbody> <tr> <td>96.0 ± 10.0 Kgm</td> <td></td> </tr> </tbody> </table> - fit O-ring (7) and install cover (6) Tightening bolt cover <table border="1"> <thead> <tr> <th>Standard</th> <th>Actual</th> </tr> </thead> <tbody> <tr> <td>11.5 ± 0.5 Kgm</td> <td></td> </tr> </tbody> </table> - Install final drive assembly (4) to housing (2) Tightening bolt (3) <table border="1"> <thead> <tr> <th>Standard</th> <th>Actual</th> </tr> </thead> <tbody> <tr> <td>135 ± 15 Kgm</td> <td></td> </tr> </tbody> </table>	Standard	Actual	5.9 - 6.9 Kgm		Actual		Standard	Actual	56.0 ± 6.0 Kgm		Standard	Actual	96.0 ± 10.0 Kgm		Standard	Actual	11.5 ± 0.5 Kgm		Standard	Actual	135 ± 15 Kgm				   
Standard	Actual																										
5.9 - 6.9 Kgm																											
Actual																											
Standard	Actual																										
56.0 ± 6.0 Kgm																											
Standard	Actual																										
96.0 ± 10.0 Kgm																											
Standard	Actual																										
11.5 ± 0.5 Kgm																											
Standard	Actual																										
135 ± 15 Kgm																											

NOTE :

INSPECTION BY

APPROVED BY

MECHANIC

GL /QA



PT SAPTAINDRA SEJATI

FINAL DRIVE RH

PERFORMANCE TEST

CHECKSHEET

MACHINE MODEL

HD785-7

RO NO :

POWERTRAIN SECTION



**QA SHEET TEST
PERFORMANCE HD785-7**

PUBLIS.DATE	10 February 2018
REVISI	0

MACHINE MODEL		PART.NO		USED OIL TO 30
COMP.MODEL		TEST DATE		
COMP.S/N		TEST BY		
WO.NO		USED OIL		

FILL OIL	STD	ACTUAL
	Liter	Liter
FINAL DRIVE RH	64	
FINAL DRIVE LH	64	
AXLE HOUSING	137	

DESCRIPTION	STD	ACTUAL		REMARK
		CCW	CW	
BLEEDING OIL BRAKE	No bubbles			Do before test running
PARKING BRAKE RELEASE PRESSURE	85 - 110 kg/cm ²			
SERVICE BRAKE ACTIVE PRESSURE	5 - 25 kg/cm ²			Apply every 10 minute rotation
BRAKE COOLING PRESSURE	1 - 6 kg/cm ²			
RPM INPUT	200 - 300 rpm			
TEMPERATURE	50 - 55 ° C			
DURATION TOTAL TIME	30 ± 10 minutes			
OIL LEAKAGE	No leak			
CHECK MAGNETIC PLUG	No particle			
OIL SAMPLE				

TEST LEAK BRAKE PRESSURE (POSITION NO ROTATION)								
BRAKE TYPE	STD PRESSURE kg/cm ²	ACTUAL MEASUREMENT						
		MINUTE	0	2	4	6	8	10
PARKING BRAKE	85 -110	kg/cm ²						
SERVICE BRAKE	85 -110	kg/cm ²						